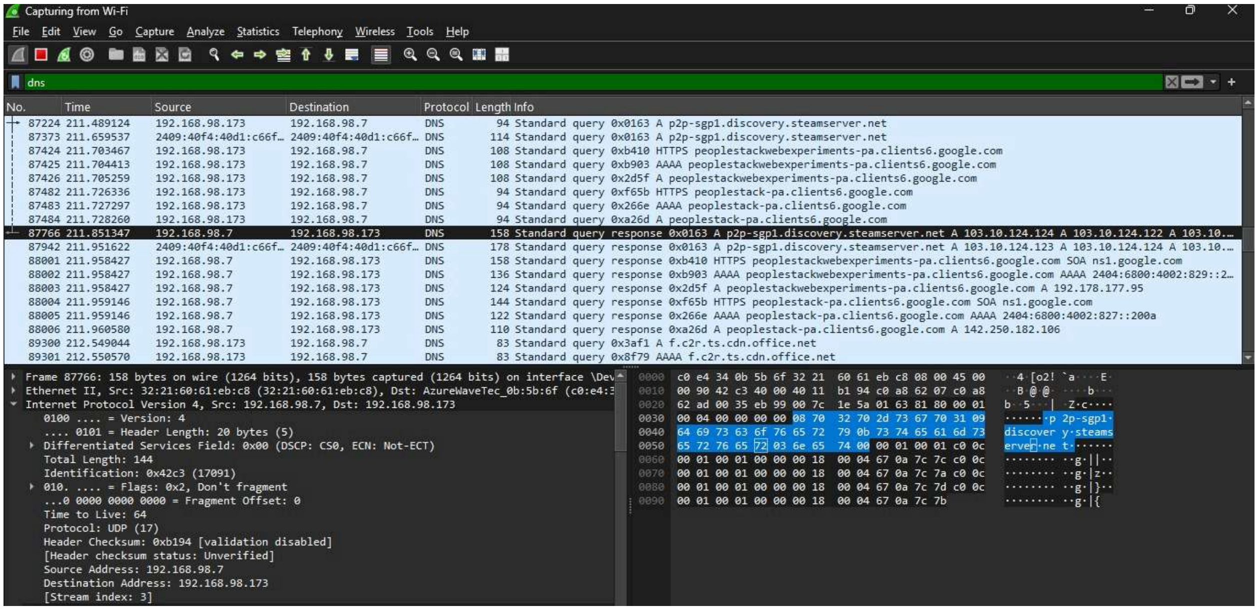
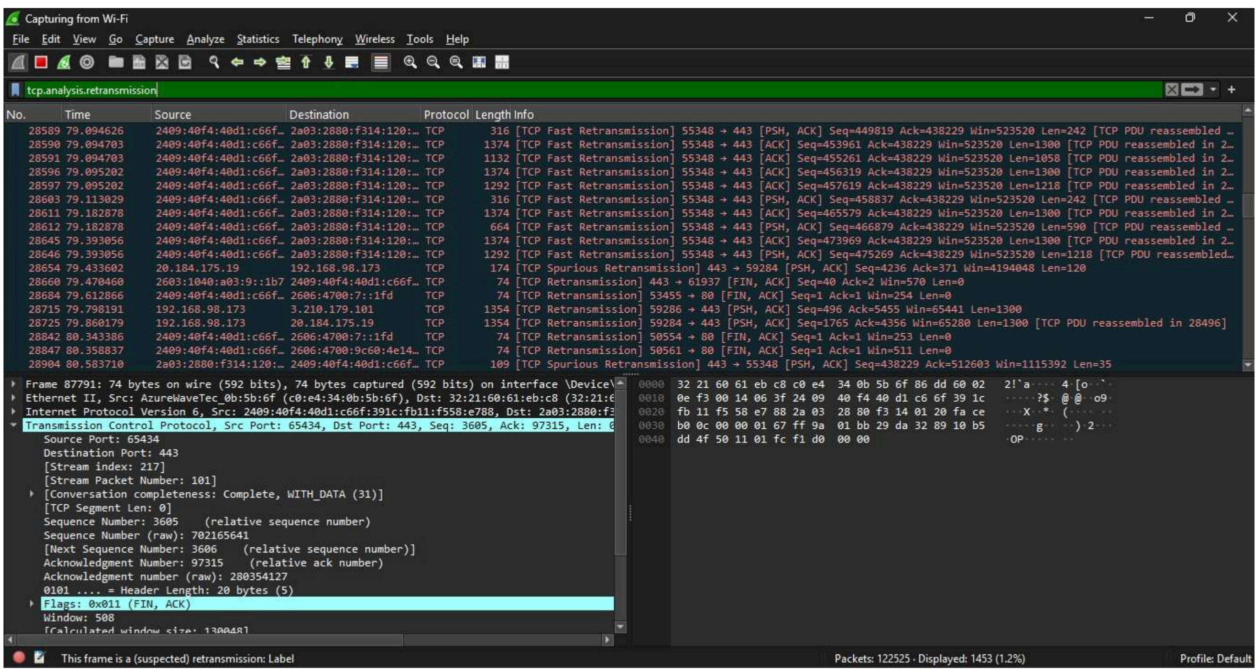
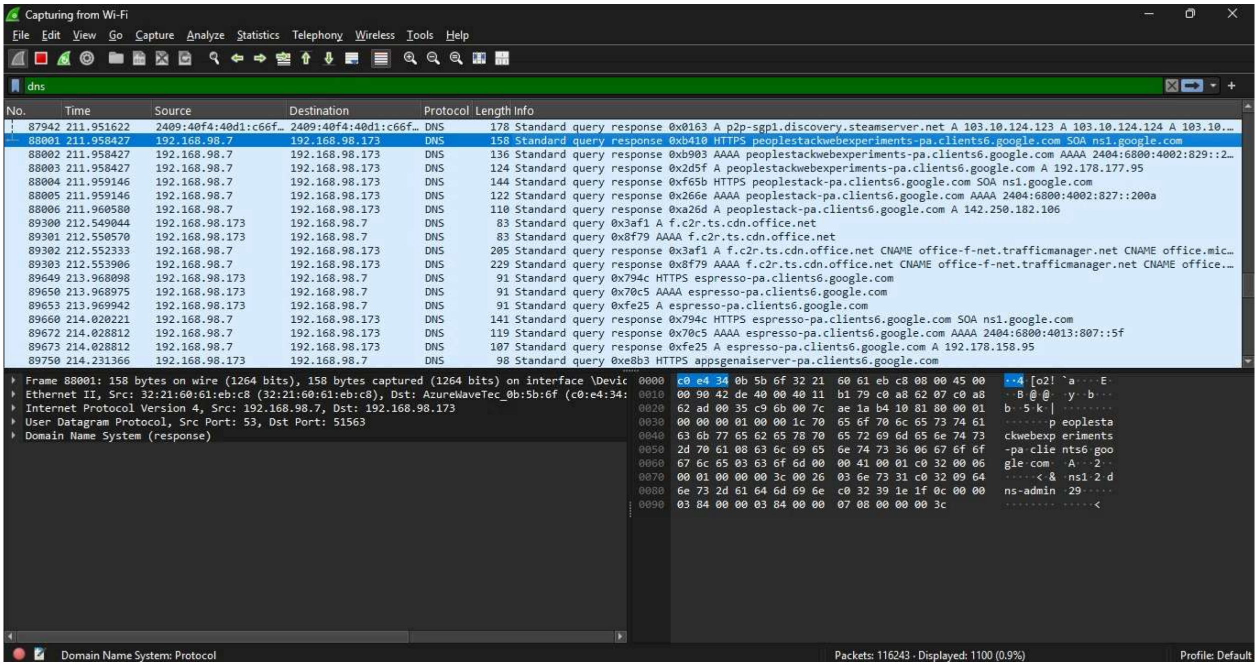
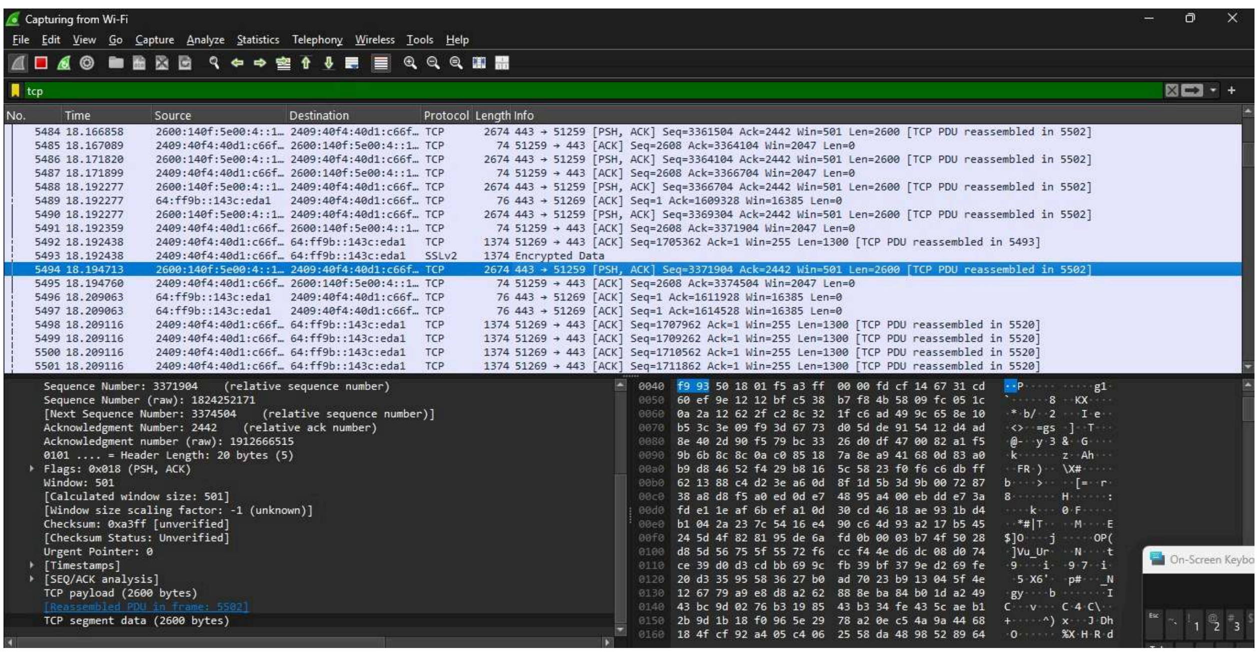

Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
87224 211.489124 192.168.98.173 192.168.98.7 DNS 94 Standard query 0x0163 A p2p-sgp1.discovery.steamserver.net
87373 211.659537 2409:40f4:40d1:c66f… 2409:40f4:40d1:c66f… DNS 114 Standard query 0x0163 A p2p-sgp1.discovery.steamserver.net
87424 211.703467 192.168.98.173 192.168.98.7 DNS 108 Standard query 0xb410 HTTPS peoplestackwebexperiments-pa.clients6.google.com
87425 211.704413 192.168.98.173 192.168.98.7 DNS 108 Standard query 0xb903 AAAA peoplestackwebexperiments-pa.clients6.google.com
87426 211.705259 192.168.98.173 192.168.98.7 DNS 108 Standard query 0x2d5f A peoplestackwebexperiments-pa.clients6.google.com
87482 211.726336 192.168.98.173 192.168.98.7 DNS 94 Standard query 0xf65b HTTPS peoplestack-pa.clients6.google.com
87483 211.727297 192.168.98.173 192.168.98.7 DNS 94 Standard query 0x266e AAAA peoplestack-pa.clients6.google.com
87484 211.728260 192.168.98.173 192.168.98.7 DNS 94 Standard query 0xa26d A peoplestack-pa.clients6.google.com
87766 211.851347 192.168.98.7 192.168.98.173 DNS 158 Standard query response 0x0163 A p2p-sgp1.discovery.steamserver.net A 103.10.124.124 A 103.10.124.122 A 103.10.…
87942 211.951622 2409:40f4:40d1:c66f… 2409:40f4:40d1:c66f… DNS 178 Standard query response 0x0163 A p2p-sgp1.discovery.steamserver.net A 103.10.124.123 A 103.10.124.124 A 103.10.…
88001 211.958427 192.168.98.7 192.168.98.173 DNS 158 Standard query response 0xb410 HTTPS peoplestackwebexperiments-pa.clients6.google.com SOA ns1.google.com
88002 211.958427 192.168.98.7 192.168.98.173 DNS 136 Standard query response 0xb903 AAAA peoplestackwebexperiments-pa.clients6.google.com AAAA 2404:6800:4002:829::2…
88003 211.958427 192.168.98.7 192.168.98.173 DNS 124 Standard query response 0x2d5f A peoplestackwebexperiments-pa.clients6.google.com A 192.178.177.95
88004 211.959146 192.168.98.7 192.168.98.173 DNS 144 Standard query response 0xf65b HTTPS peoplestack-pa.clients6.google.com SOA ns1.google.com
88005 211.959146 192.168.98.7 192.168.98.173 DNS 122 Standard query response 0x266e AAAA peoplestack-pa.clients6.google.com AAAA 2404:6800:4002:827::200a
88006 211.960580 192.168.98.7 192.168.98.173 DNS 110 Standard query response 0xa26d A peoplestack-pa.clients6.google.com A 142.250.182.106
89300 212.549044 192.168.98.173 192.168.98.7 DNS 83 Standard query 0x3af1 A f.c2r.ts.cdn.office.net
89301 212.550570 192.168.98.173 192.168.98.7 DNS 83 Standard query 0x8f79 AAAA f.c2r.ts.cdn.office.net
Frame 87766: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface \Dev
Ethernet II, Src: 32:21:60:61:eb:c8 (32:21:60:61:eb:c8), Dst: AzureWaveTec_0b:5b:6f (c0:e4:3
Internet Protocol Version 4, Src: 192.168.98.7, Dst: 192.168.98.173
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 144
Identification: 0x42c3 (17091)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 64
Protocol: UDP (17)
Header Checksum: 0xb194 [validation disabled]
[Header checksum status: Unverified]
Source Address: 192.168.98.7
Destination Address: 192.168.98.173
[Stream index: 3]
0000 c0 e4 34 0b 5b 6f 32 21 60 61 eb c8 08 00 45 00
0010 00 90 42 c3 40 00 40 11 b1 94 c0 a8 62 07 c0 a8
0020 62 ad 00 35 eb 99 00 7c 1e 5a 01 63 81 80 00 01
0030 00 04 00 00 00 00 08 70 32 70 2d 73 67 70 31 09
0040 64 69 73 63 6f 76 65 72 79 0b 73 74 65 61 6d 73
0050 65 72 76 65 72 03 6e 65 74 00 00 01 00 01 c0 0c
0060 00 01 00 01 00 00 00 18 00 04 67 0a 7c 7c c0 0c
0070 00 01 00 01 00 00 00 18 00 04 67 0a 7c 7a c0 0c
0080 00 01 00 01 00 00 00 18 00 04 67 0a 7c 7d c0 0c
0090 00 01 00 01 00 00 00 18 00 04 67 0a 7c 7b


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.analysis.retransmission
No. Time Source Destination Protocol Length Info
28589 79.094626 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 316 [TCP Fast Retransmission] 55348 → 443 [PSH, ACK] Seq=449819 Ack=438229 Win=523520 Len=242 [TCP PDU reassembled …
28590 79.094703 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1374 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=453961 Ack=438229 Win=523520 Len=1300 [TCP PDU reassembled in 2…
28591 79.094703 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1132 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=455261 Ack=438229 Win=523520 Len=1058 [TCP PDU reassembled in 2…
28596 79.095202 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1374 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=456319 Ack=438229 Win=523520 Len=1300 [TCP PDU reassembled in 2…
28597 79.095202 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1292 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=457619 Ack=438229 Win=523520 Len=1218 [TCP PDU reassembled in 2…
28603 79.113029 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 316 [TCP Fast Retransmission] 55348 → 443 [PSH, ACK] Seq=458837 Ack=438229 Win=523520 Len=242 [TCP PDU reassembled …
28611 79.182878 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1374 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=465579 Ack=438229 Win=523520 Len=1300 [TCP PDU reassembled in 2…
28612 79.182878 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 664 [TCP Fast Retransmission] 55348 → 443 [PSH, ACK] Seq=466879 Ack=438229 Win=523520 Len=590 [TCP PDU reassembled …
28645 79.393056 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1374 [TCP Fast Retransmission] 55348 → 443 [ACK] Seq=473969 Ack=438229 Win=523520 Len=1300 [TCP PDU reassembled in 2…
28646 79.393056 2409:40f4:40d1:c66f… 2a03:2880:f314:120:… TCP 1292 [TCP Fast Retransmission] 55348 → 443 [PSH, ACK] Seq=475269 Ack=438229 Win=523520 Len=1218 [TCP PDU reassembled…
28654 79.433602 20.184.175.19 192.168.98.173 TCP 174 [TCP Spurious Retransmission] 443 → 59284 [PSH, ACK] Seq=4236 Ack=371 Win=4194048 Len=120
28660 79.470460 2603:1040:a03:9::1b7 2409:40f4:40d1:c66f… TCP 74 [TCP Retransmission] 443 → 61937 [FIN, ACK] Seq=40 Ack=2 Win=570 Len=0
28684 79.612866 2409:40f4:40d1:c66f… 2606:4700:7::1fd TCP 74 [TCP Retransmission] 53455 → 80 [FIN, ACK] Seq=1 Ack=1 Win=254 Len=0
28715 79.798191 192.168.98.173 3.210.179.101 TCP 1354 [TCP Retransmission] 59286 → 443 [PSH, ACK] Seq=496 Ack=5455 Win=65441 Len=1300
28725 79.860179 192.168.98.173 20.184.175.19 TCP 1354 [TCP Retransmission] 59284 → 443 [PSH, ACK] Seq=1765 Ack=4356 Win=65280 Len=1300 [TCP PDU reassembled in 28496]
28842 80.343386 2409:40f4:40d1:c66f… 2606:4700:7::1fd TCP 74 [TCP Retransmission] 50554 → 80 [FIN, ACK] Seq=1 Ack=1 Win=253 Len=0
28847 80.358837 2409:40f4:40d1:c66f… 2606:4700:9c60:4e14… TCP 74 [TCP Retransmission] 50561 → 80 [FIN, ACK] Seq=1 Ack=1 Win=511 Len=0
28904 80.583710 2a03:2880:f314:120:… 2409:40f4:40d1:c66f… TCP 109 [TCP Spurious Retransmission] 443 → 55348 [PSH, ACK] Seq=438229 Ack=512603 Win=1115392 Len=35
Frame 87791: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface \Device\
Ethernet II, Src: AzureWaveTec_0b:5b:6f (c0:e4:34:0b:5b:6f), Dst: 32:21:60:61:eb:c8 (32:21:6
Internet Protocol Version 6, Src: 2409:40f4:40d1:c66f:391c:fb11:f558:e788, Dst: 2a03:2880:f3
Transmission Control Protocol, Src Port: 65434, Dst Port: 443, Seq: 3605, Ack: 97315, Len: 0
Source Port: 65434
Destination Port: 443
[Stream index: 217]
[Stream Packet Number: 101]
[Conversation completeness: Complete, WITH_DATA (31)]
[TCP Segment Len: 0]
Sequence Number: 3605 (relative sequence number)
Sequence Number (raw): 702165641
[Next Sequence Number: 3606 (relative sequence number)]
Acknowledgment Number: 97315 (relative ack number)
Acknowledgment number (raw): 280354127
0101 .... = Header Length: 20 bytes (5)
Flags: 0x011 (FIN, ACK)
Window: 508
0000 32 21 60 61 eb c8 c0 e4 34 0b 5b 6f 86 dd 60 02
0010 0e f3 00 14 06 3f 24 09 40 f4 40 d1 c6 6f 39 1c
0020 fb 11 f5 58 e7 88 2a 03 28 80 f3 14 01 20 fa ce
0030 b0 0c 00 00 01 67 ff 9a 01 bb 29 da 32 89 10 b5
0040 dd 4f 50 11 01 fc f1 d0 00 00
This frame is a (suspected) retransmission: Label
Packets: 122525 · Displayed: 1453 (1.2%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
87942 211.951622 2409:40f4:40d1:c66f… 2409:40f4:40d1:c66f… DNS 178 Standard query response 0x0163 A p2p-sgp1.discovery.steamserver.net A 103.10.124.123 A 103.10.124.124 A 103.10…
88001 211.958427 192.168.98.7 192.168.98.173 DNS 158 Standard query response 0xb410 HTTPS peoplestackwebexperiments-pa.clients6.google.com SOA ns1.google.com
88002 211.958427 192.168.98.7 192.168.98.173 DNS 136 Standard query response 0xb903 AAAA peoplestackwebexperiments-pa.clients6.google.com AAAA 2404:6800:4002:829::2…
88003 211.958427 192.168.98.7 192.168.98.173 DNS 124 Standard query response 0x2d5f A peoplestackwebexperiments-pa.clients6.google.com A 192.178.177.95
88004 211.959146 192.168.98.7 192.168.98.173 DNS 144 Standard query response 0xf65b HTTPS peoplestack-pa.clients6.google.com SOA ns1.google.com
88005 211.959146 192.168.98.7 192.168.98.173 DNS 122 Standard query response 0x266e AAAA peoplestack-pa.clients6.google.com AAAA 2404:6800:4002:827::200a
88006 211.960580 192.168.98.7 192.168.98.173 DNS 110 Standard query response 0xa26d A peoplestack-pa.clients6.google.com A 142.250.182.106
89300 212.549044 192.168.98.173 192.168.98.7 DNS 83 Standard query 0x3af1 A f.c2r.ts.cdn.office.net
89301 212.550570 192.168.98.173 192.168.98.7 DNS 83 Standard query 0x8f79 AAAA f.c2r.ts.cdn.office.net
89302 212.552333 192.168.98.7 192.168.98.173 DNS 205 Standard query response 0x3af1 A f.c2r.ts.cdn.office.net CNAME office-f-net.trafficmanager.net CNAME office.mic…
89303 212.553906 192.168.98.7 192.168.98.173 DNS 229 Standard query response 0x8f79 AAAA f.c2r.ts.cdn.office.net CNAME office-f-net.trafficmanager.net CNAME office.…
89649 213.968098 192.168.98.173 192.168.98.7 DNS 91 Standard query 0x794c HTTPS espresso-pa.clients6.google.com
89650 213.968975 192.168.98.173 192.168.98.7 DNS 91 Standard query 0x70c5 AAAA espresso-pa.clients6.google.com
89653 213.969942 192.168.98.173 192.168.98.7 DNS 91 Standard query 0xfe25 A espresso-pa.clients6.google.com
89660 214.020221 192.168.98.7 192.168.98.173 DNS 141 Standard query response 0x794c HTTPS espresso-pa.clients6.google.com SOA ns1.google.com
89672 214.028812 192.168.98.7 192.168.98.173 DNS 119 Standard query response 0x70c5 AAAA espresso-pa.clients6.google.com AAAA 2404:6800:4013:807::5f
89673 214.028812 192.168.98.7 192.168.98.173 DNS 107 Standard query response 0xfe25 A espresso-pa.clients6.google.com A 192.178.158.95
89750 214.231366 192.168.98.173 192.168.98.7 DNS 98 Standard query 0xe8b3 HTTPS appsgenaiserver-pa.clients6.google.com
Frame 88001: 158 bytes on wire (1264 bits), 158 bytes captured (1264 bits) on interface \Devic
Ethernet II, Src: 32:21:60:61:eb:c8 (32:21:60:61:eb:c8), Dst: AzureWaveTec_0b:5b:6f (c0:e4:34:
Internet Protocol Version 4, Src: 192.168.98.7, Dst: 192.168.98.173
User Datagram Protocol, Src Port: 53, Dst Port: 51563
Domain Name System (response)
0000 c0 e4 34 0b 5b 6f 32 21 60 61 eb c8 08 00 45 00
0010 00 90 42 de 40 00 40 11 b1 79 c0 a8 62 07 c0 a8
0020 62 ad 00 35 c9 6b 00 7c ae 1a b4 10 81 80 00 01
0030 00 00 00 01 00 00 1c 70 65 6f 70 6c 65 73 74 61
0040 63 6b 77 65 62 65 78 70 65 72 69 6d 65 6e 74 73
0050 2d 70 61 08 63 6c 69 65 6e 74 73 36 06 67 6f 6f
0060 67 6c 65 03 63 6f 6d 00 00 41 00 01 c0 32 00 06
0070 00 01 00 00 00 3c 00 26 03 6e 73 31 c0 32 09 64
0080 6e 73 2d 61 64 6d 69 6e c0 32 39 1e 1f 0c 00 00
0090 03 84 00 00 03 84 00 00 07 08 00 00 00 3c
Domain Name System: Protocol
Packets: 116243 · Displayed: 1100 (0.9%)
Profile: Default


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp
No. Time Source Destination Protocol Length Info
5484 18.166858 2600:140f:5e00:4::1… 2409:40f4:40d1:c66f… TCP 2674 443 → 51259 [PSH, ACK] Seq=3361504 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5485 18.167089 2409:40f4:40d1:c66f… 2600:140f:5e00:4::1… TCP 74 51259 → 443 [ACK] Seq=2608 Ack=3364104 Win=2047 Len=0
5486 18.171820 2600:140f:5e00:4::1… 2409:40f4:40d1:c66f… TCP 2674 443 → 51259 [PSH, ACK] Seq=3364104 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5487 18.171899 2409:40f4:40d1:c66f… 2600:140f:5e00:4::1… TCP 74 51259 → 443 [ACK] Seq=2608 Ack=3366704 Win=2047 Len=0
5488 18.192277 2600:140f:5e00:4::1… 2409:40f4:40d1:c66f… TCP 2674 443 → 51259 [PSH, ACK] Seq=3366704 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5489 18.192277 64:ff9b::143c:eda1 2409:40f4:40d1:c66f… TCP 76 443 → 51269 [ACK] Seq=1 Ack=1609328 Win=16385 Len=0
5490 18.192277 2600:140f:5e00:4::1… 2409:40f4:40d1:c66f… TCP 2674 443 → 51259 [PSH, ACK] Seq=3369304 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5491 18.192359 2409:40f4:40d1:c66f… 2600:140f:5e00:4::1… TCP 74 51259 → 443 [ACK] Seq=2608 Ack=3371904 Win=2047 Len=0
5492 18.192438 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 TCP 1374 51269 → 443 [ACK] Seq=1705362 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5493]
5493 18.192438 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 SSLv2 1374 Encrypted Data
5494 18.194713 2600:140f:5e00:4::1… 2409:40f4:40d1:c66f… TCP 2674 443 → 51259 [PSH, ACK] Seq=3371904 Ack=2442 Win=501 Len=2600 [TCP PDU reassembled in 5502]
5495 18.194760 2409:40f4:40d1:c66f… 2600:140f:5e00:4::1… TCP 74 51259 → 443 [ACK] Seq=2608 Ack=3374504 Win=2047 Len=0
5496 18.209063 64:ff9b::143c:eda1 2409:40f4:40d1:c66f… TCP 76 443 → 51269 [ACK] Seq=1 Ack=1611928 Win=16385 Len=0
5497 18.209063 64:ff9b::143c:eda1 2409:40f4:40d1:c66f… TCP 76 443 → 51269 [ACK] Seq=1 Ack=1614528 Win=16385 Len=0
5498 18.209116 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 TCP 1374 51269 → 443 [ACK] Seq=1707962 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5499 18.209116 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 TCP 1374 51269 → 443 [ACK] Seq=1709262 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5500 18.209116 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 TCP 1374 51269 → 443 [ACK] Seq=1710562 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
5501 18.209116 2409:40f4:40d1:c66f… 64:ff9b::143c:eda1 TCP 1374 51269 → 443 [ACK] Seq=1711862 Ack=1 Win=255 Len=1300 [TCP PDU reassembled in 5520]
Sequence Number: 3371904 (relative sequence number)
Sequence Number (raw): 1824252171
[Next Sequence Number: 3374504 (relative sequence number)]
Acknowledgment Number: 2442 (relative ack number)
Acknowledgment number (raw): 1912666515
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 501
[Calculated window size: 501]
[Window size scaling factor: -1 (unknown)]
Checksum: 0xa3ff [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
TCP payload (2600 bytes)
[Reassembled PDU in frame: 5502]
TCP segment data (2600 bytes)
On-Screen Keybo

SIMATS ENGINEERING
ASSESSMENT TOOL 1
Experiments

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Code of Conduct:

I <u>Sree B (192524023)</u> (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sree B

Experiments

QUESTIONS:4 **Total Marks:40**

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation